DIGITAL IMAGE PROCESSING (DIP)

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Section 2: Color Representation

Fall 1405
Fall 2026

Outline

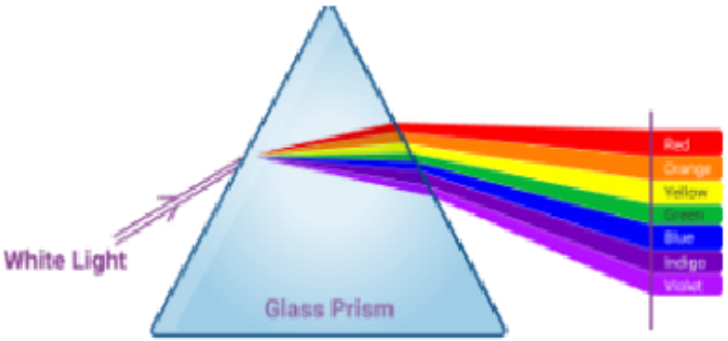
MATLAB or Python may be used to perform simulations and present the results. Students are free to use either software or both.

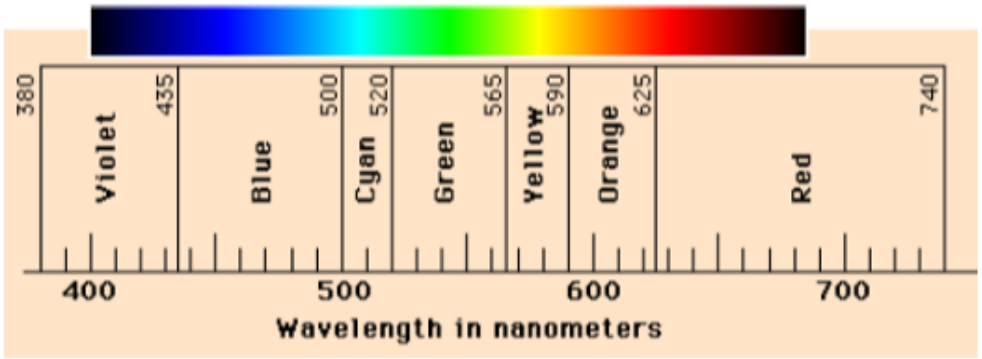
- RGB Space: Red-Green-Blue
- HSV Space: Hue-Saturation-Value
- Lab Color Space
- YCbCr Color Space
- Additive VS. Subtractive Color Mixing
- Applications

Color Definition

Light has a spectral power distribution; the visual system responds to it; and color is the resulting perceptual experience. COLOR is an important feature:

Human can distinguish thousands of color intensities and shades.





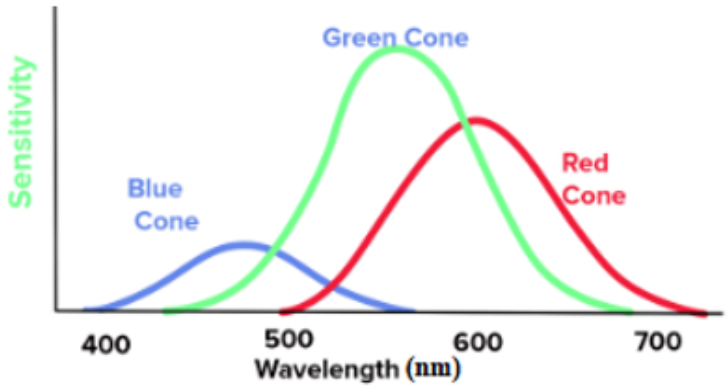
Blue 435.8 nm | Green 546.1 nm | Red 700 nm

Spectral energy distribution of incident light $C(\lambda)$

$$\int_\lambda R(\lambda)C(\lambda)d\lambda \rightarrow \alpha_R$$
$$\int_\lambda G(\lambda)C(\lambda)d\lambda \rightarrow \alpha_G$$
$$\int_\lambda B(\lambda)C(\lambda)d\lambda \rightarrow \alpha_B$$

Effective cone stimulation

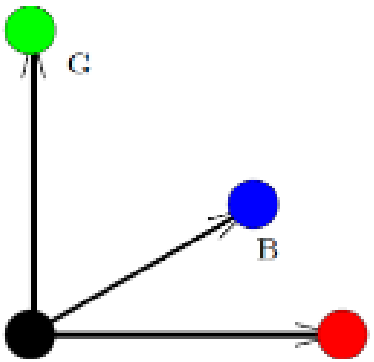
(tristimulus Values)



RGB Space

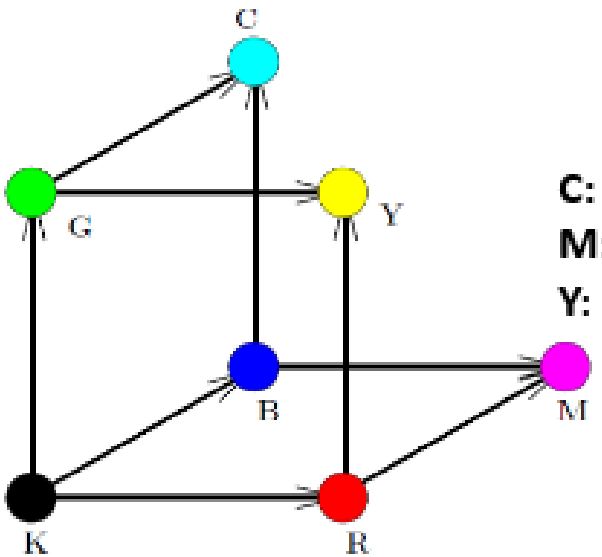
Color Model: Color Space

RGB: RED, GREEN, BLUE

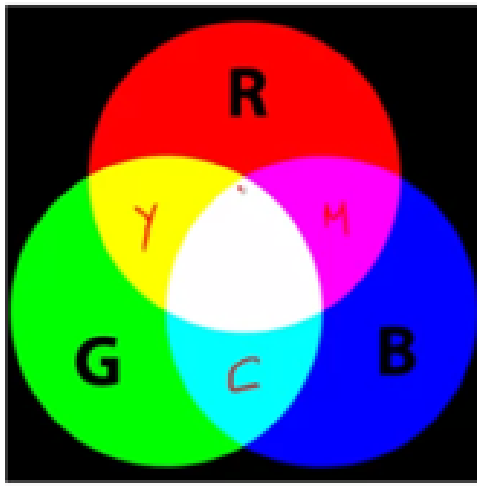


$$\text{Color} = r\,R + g\,G + b\,B = \begin{bmatrix} r & g & b \end{bmatrix} \begin{bmatrix} R \\ G \\ B \end{bmatrix}$$

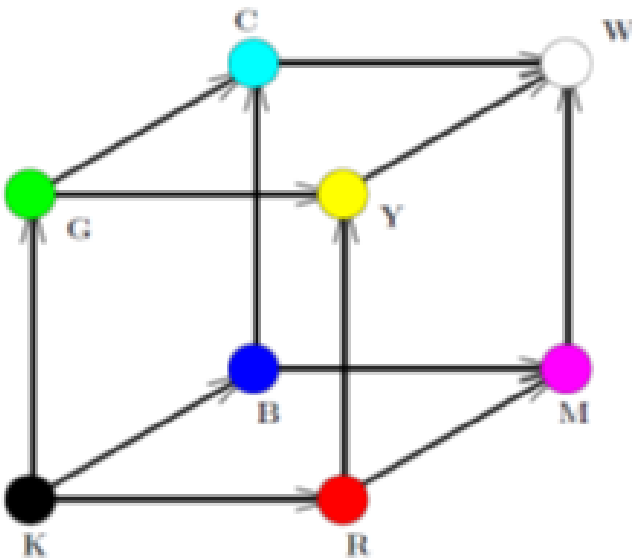
Red = R $\Leftrightarrow$ (r, g, b) = (1, 0, 0)
Green = G $\Leftrightarrow$ (r, g, b) = (0, 1, 0)
Blue = B $\Leftrightarrow$ (r, g, b) = (0, 0, 1)



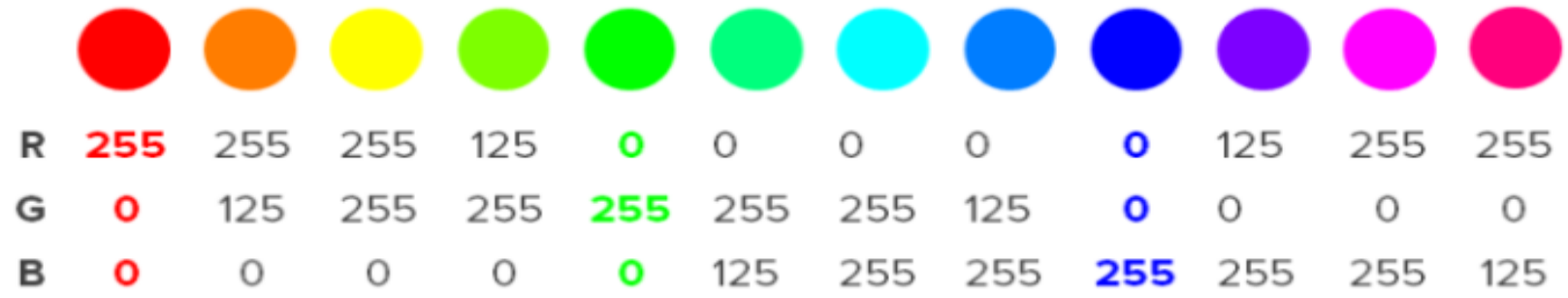
C: Cyan: فیروزه ای
M: Magenta: سرخابی
Y: Yellow: زرد



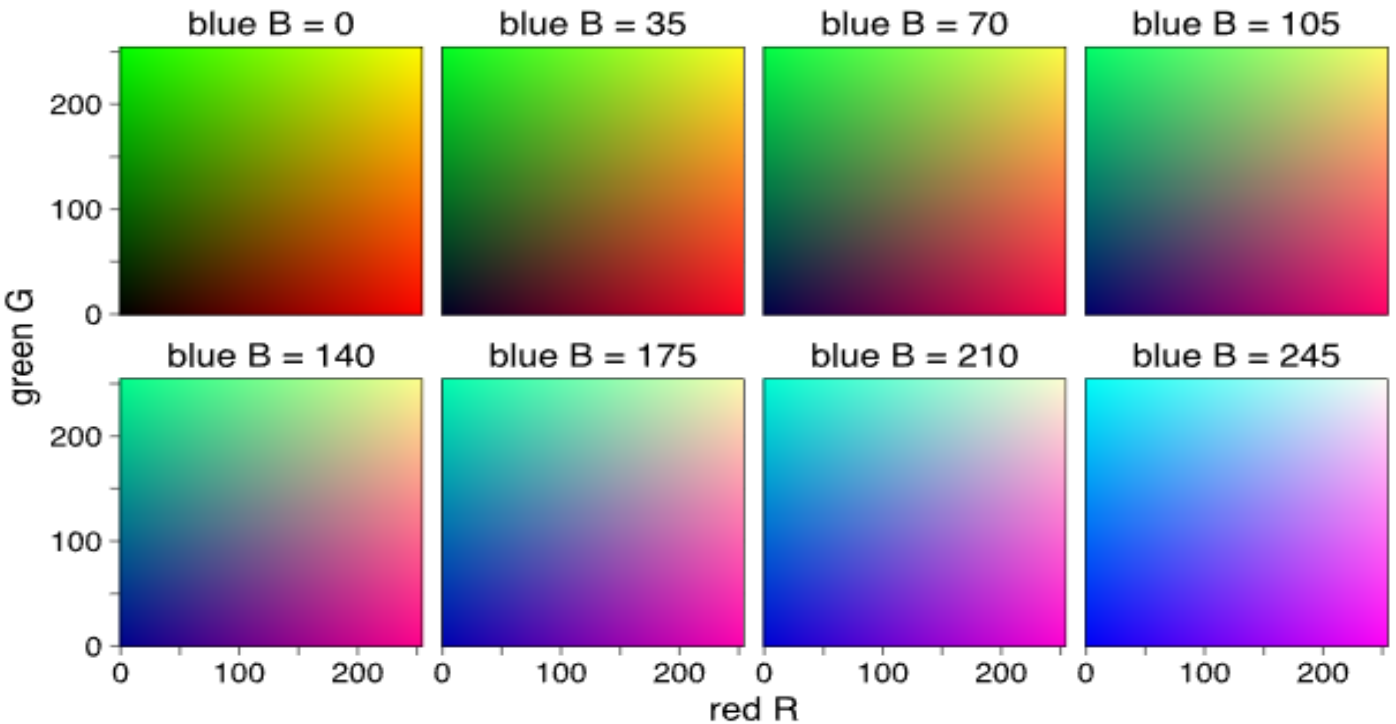
additive



RGB Space: continued



R	255	255	255	125	0	0	0	0	0	125	255	255
G	0	125	255	255	255	255	255	125	0	0	0	0
B	0	0	0	0	0	125	255	255	255	255	255	125



Example:

RGB



Red channel



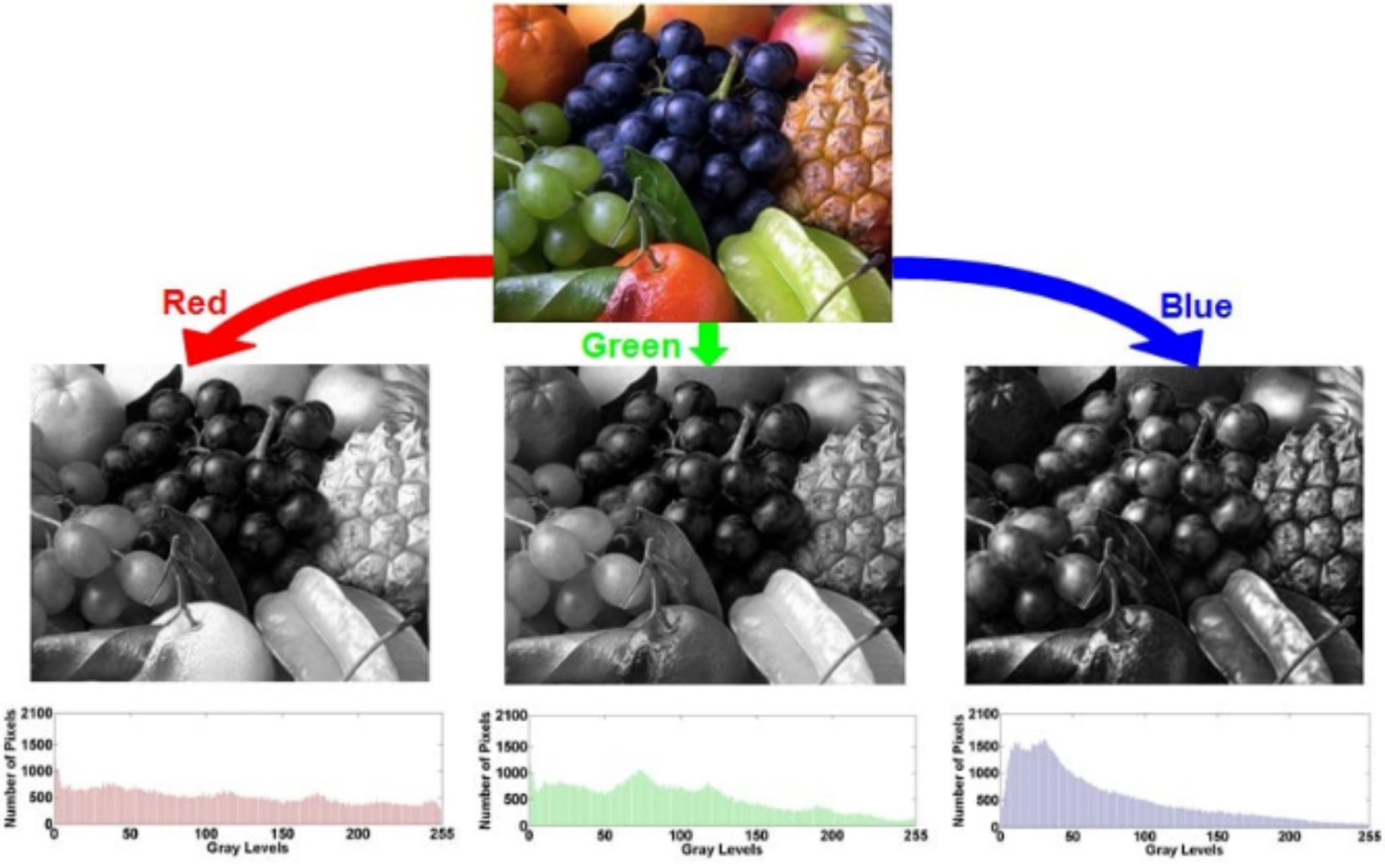
Green channel



Blue channel



Example:


Red
Green
Blue
Number of Pixels
2100
1500
1000
500
0
0
50
100
150
200
255
Gray Levels

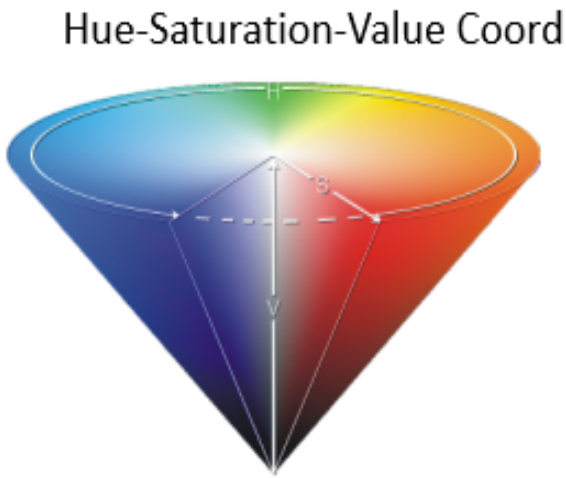
HSV Space

Hue: [0: 360] Degree فام

Saturation: [0:100] خلوص

Value-Illumination: روشنائی
Brightness [0:100] انرژی

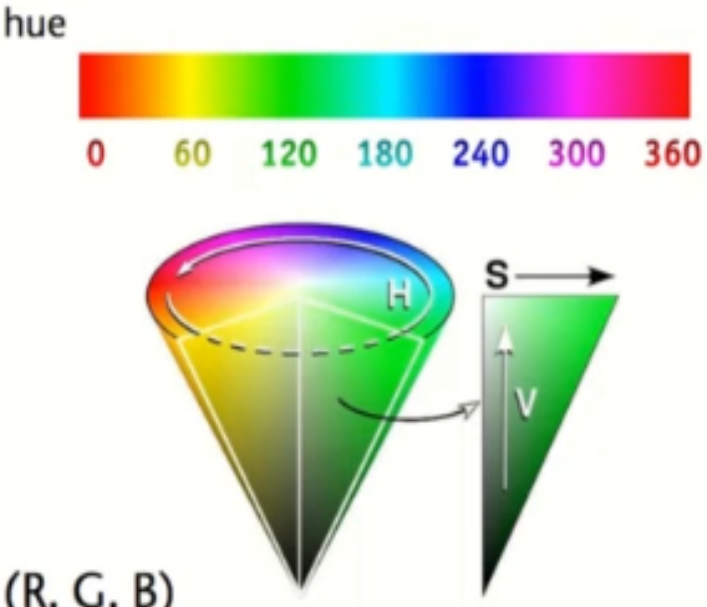








RGB → HSV Convertion



- colour cone
  - H = hue / colour in degrees ∈ [0,360]
  - S = saturation ∈ [0,1]
  - V = value ∈ [0,1]
- conversion RGB → HSV
  - V = max = max (R, G, B), min = min (R, G, B)
  - S = (max – min) / max (or S = 0, if V = 0)
  - $$H = 60 \times \begin{cases} 0 + (G - B)/(\max - \min), & \text{if } \max = R \\ 2 + (B - R)/(\max - \min), & \text{if } \max = G \\ 4 + (R - G)/(\max - \min), & \text{if } \max = B \end{cases}$$

    $H = H + 360$, if $H < 0$

HSV $\rightarrow$ RGB

Given $H \in [0°, 360°)$, $S, V \in [0, 1]$:

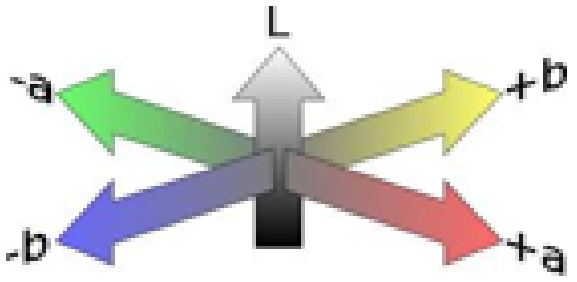
$$C = V\,S, \qquad H' = \frac{H}{60°},$$
$$X = C\left(1 - \left|\, H' \bmod 2 - 1 \right|\right), \qquad m = V - C$$

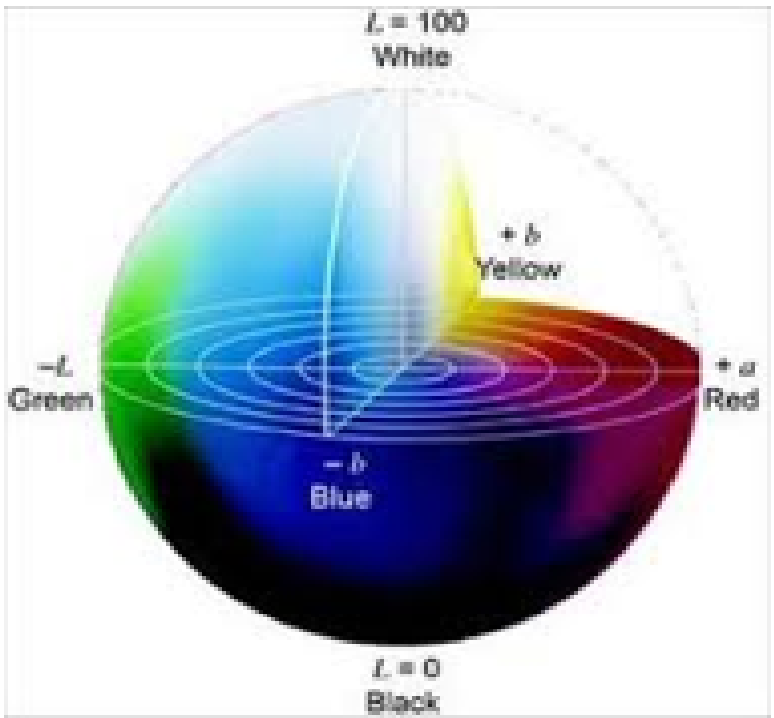
$$(R_1, G_1, B_1) = \begin{cases} (C, X, 0), & 0 \le H' < 1 \\ (X, C, 0), & 1 \le H' < 2 \\ (0, C, X), & 2 \le H' < 3 \\ (0, X, C), & 3 \le H' < 4 \\ (X, 0, C), & 4 \le H' < 5 \\ (C, 0, X), & 5 \le H' < 6 \end{cases}$$

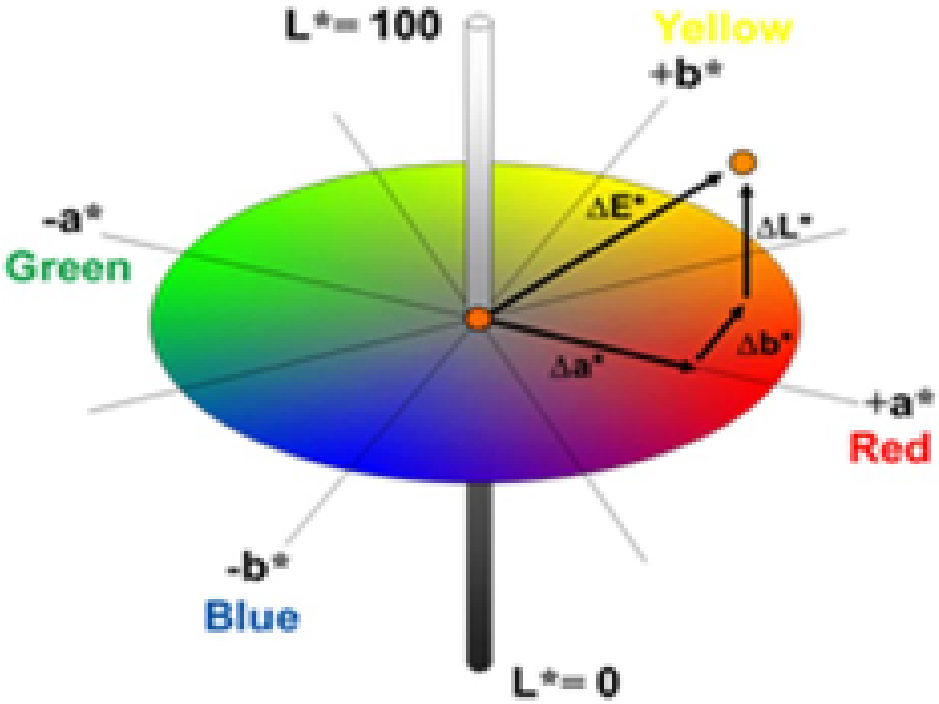
$$(R, G, B) = (R_1 + m,\ G_1 + m,\ B_1 + m)$$

CIELab Space

Lab defines color in terms : perceptual lightness (L*), redness-to-greenness (a*), blueness-to-yellowness (b*).







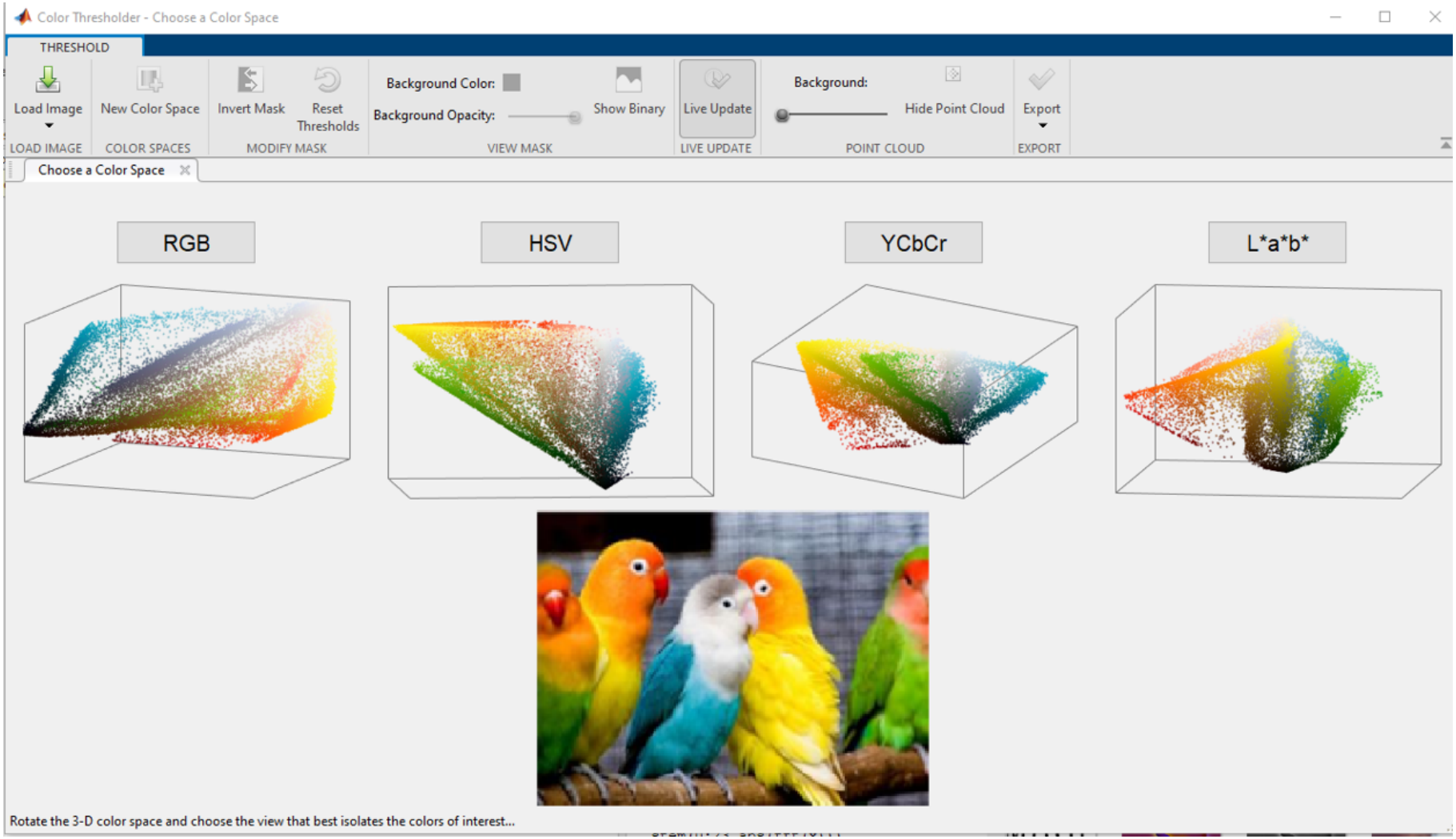
color-difference measure: $\Delta E = \sqrt{(\Delta L)^2 + (\Delta a)^2 + (\Delta b)^2}$

YCbCr Space

- Y describes the "Luminance", "Cb" and "Cr" are the chrominance.
- YUV is used by the PAL, NTSC, SECAM for color analog.
- YCbCr is discrete case of YUV, a scaled and offset version.
- JPEG uses YCbCr.
- R, G and B values in range $[0, 255]$ and Y, Cb and Cr are in range $[o, 255]$

$$\begin{bmatrix} Y \\ Cb \\ Cr \end{bmatrix} = \begin{bmatrix} 0.299 & 0.587 & 0.114 \\ -0.168736 & -0.331264 & 0.5 \\ 0.5 & -0.418688 & -0.081312 \end{bmatrix} \begin{bmatrix} R \\ G \\ B \end{bmatrix} + \begin{bmatrix} 0 \\ 128 \\ 128 \end{bmatrix}$$

Color Thresholder Application in Matlab



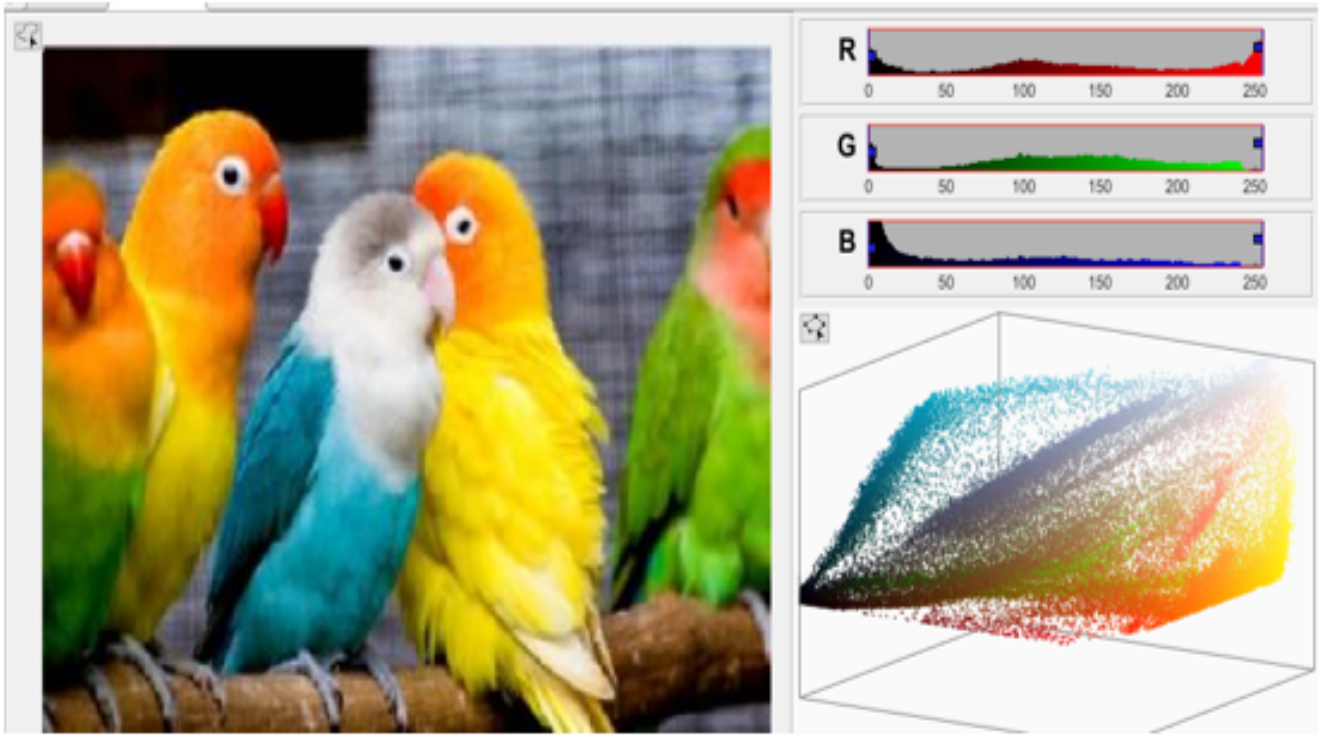
Images in RGB Components







```
m=colormap
mr=[m(:,1),zeros(256,2)];
imshow(flies(:,:,1),mr)
mg=[zeros(256,1),m(:,2),zeros(256,1)];
imshow(flies(:,:,2),mg)
mb=[zeros(256,2),m(:,3)];
```



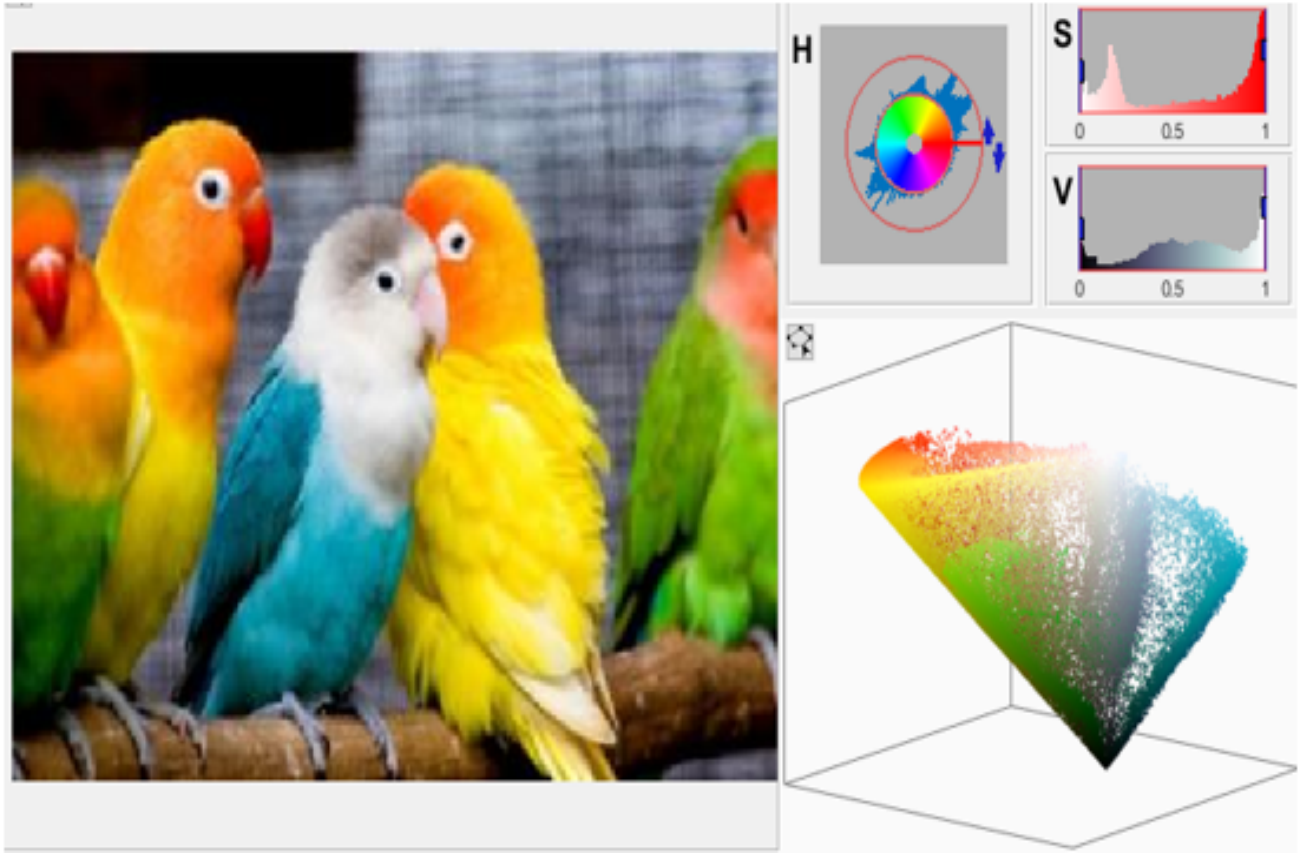
Images in HSV Components







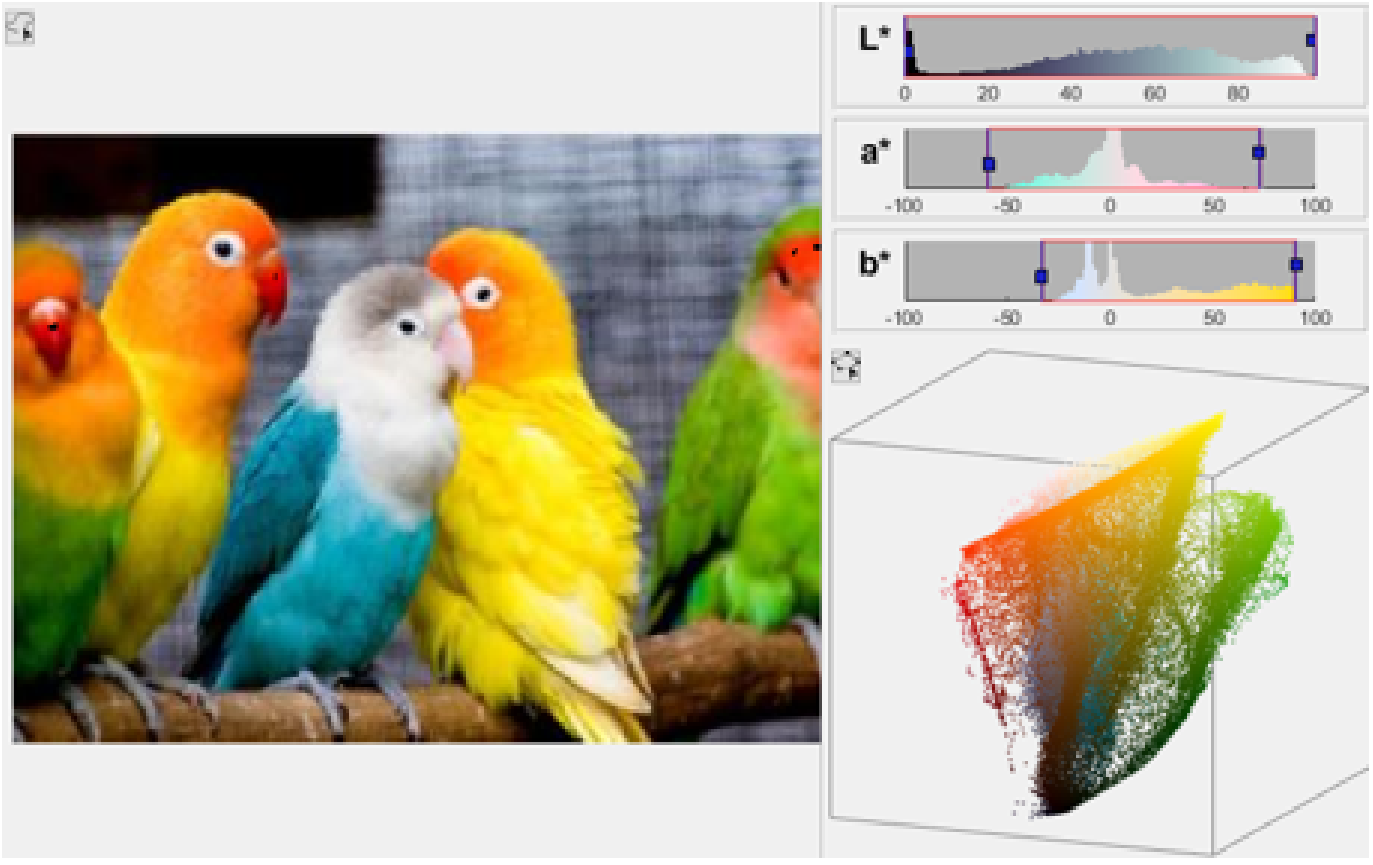
```
fhsv=rgb2hsv(flies);
imshow(fhsv(:,:,1),[])
imshow(fhsv(:,:,2),[])
imshow(fhsv(:,:,3),[])
```



Images in Lab Components



```
[L A B]=imsplit(rgb2lab(flies));
Lfill = 50*ones(size(L)); % pick some L level for representing A,B
Zfill = zeros(size(L));        Lvis = mat2gray(L);
Avis = lab2rgb(cat(3,Lfill,A,Zfill));      Bvis = lab2rgb(cat(3,Lfill,Zfill,B));
```



Images in YCbCr Components

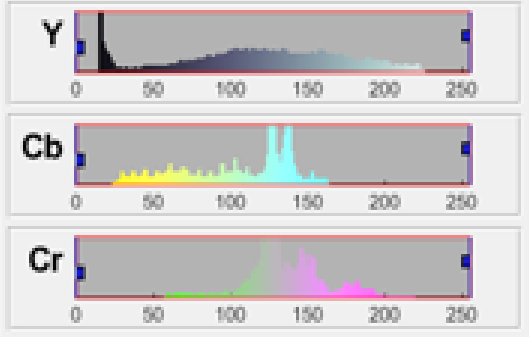


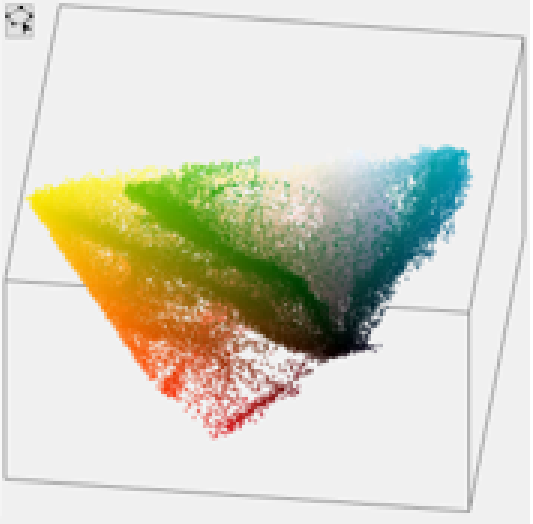




```
[Y Cb Cr] = imsplit(rgb2ycbcr(flies));
z = 127*ones(size(Y)); Ysub = 127*ones(size(Y));
just_Cb = ycbcr2rgb(cat(3, Ysub, Cb, z));
just_Cr = ycbcr2rgb(cat(3, Ysub, z, Cr));
```







Gray images in comparison

L: Lab



V: HSV



Y: YCbCr



rgb2gray(flies)



Example:



Original



R



G



B



Y



Cb



Cr

Additive VS. Subtractive Color Mixing









$$\begin{bmatrix} C \\ M \\ Y \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} - \begin{bmatrix} R \\ G \\ B \end{bmatrix}$$

CMYK

Equal amounts of C, M, Y should produce black, but they produce instead in practice a muddy-looking black. Therefore, a fourth color "black" K is added to produce the CMYK color model

Additive color is a method to create color by mixing a number of different light colors, with shades of red, green, and blue being the most common primary colors used in additive color systems.

Subtractive color is a method to create color by mixing a number of different sets of dyes, inks, paint pigments, or natural colorants to create a wider range of colors, each the result of partially or completely subtracting (that is, absorbing) some wavelengths of light and not others. A surface's color depends on which parts of the visible spectrum are not absorbed and therefore remain visible.}

Color Spaces: Concept and Applications

Space	Concept	Limitation
RGB	Additive, device-dependent; R, G, B are the sensor/display primaries.	Channels are highly correlated; not perceptual; brightness and color are mixed.
HSV	Hue, Saturation, Value; hexcone model with $V = \max(R, G, B)$, so image dependent.	Hue unstable at low S or low V; V is not perceived brightness.
CIELAB	L^* (lightness), a^*, b^* (opponent axes); device-independent, approximately perceptually uniform.	Needs a reference white point; nonlinear; conversion cost; out-of-gamut values.
YCbCr	Luma Y plus two chroma differences C_b, C_r, derived from gamma-corrected RGB.	Not perceptually uniform; coefficients depend on the standard (BT.601, BT.709).

Which Space for Which Task?

Task	**Best choice**	**Reason**
Acquisition (Cameras, sensors), display	RGB	Native sensor/display format
Enhance brightness, keep color unchanged, graphics software, thresholding by hue, skin detection, object tracking	HSV	Brightness is decoupled from color
Measuring perceived color difference, color matching and calibration, perceptual accuracy	CIELAB	ΔE is approximately Euclidean
Image/video compression	YCbCr	Chroma can be subsampled with little visible loss

Chroma Interesting Keying

The blue screen is used as a background for chroma keying in video or film production. Chroma keying is the process of replacing a solid-color background (often blue or green) with another image or video. The blue screen is used because it is the farthest color from human skin tones, making it easier to separate the subject from the background. This technique allows filmmakers to create special effects and composite different shots together.

Forrest Gump movie 1994





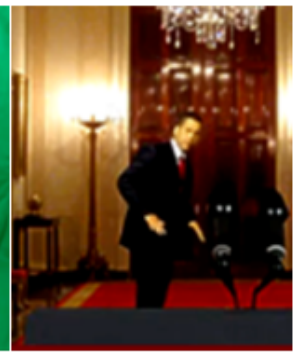












Color Gamut

- usually **at least 256** shades of each red, green and blue are employed resulting in at least $256^3 = 16,777,216$ **(16 million)** color variations.
- Human eye can discern as many as **ten million** colors.
- The human eye is most sensitive to green, then red, and last blue.
- A color with more green is brighter to the eye than a color with more blue

$$x = \frac{R}{R+G+B}; y = \frac{G}{R+G+B}; z = \frac{B}{R+G+B}$$

$$z = 1 - (x + y)$$

